

# Summary of Differences Between Adiabatic and Isothermal Processes:

| Process                | Key Characteristics                                     | Temperature Behavior  | Heat Transfer | Work Done                   |
|------------------------|---------------------------------------------------------|-----------------------|---------------|-----------------------------|
| Adiabatic Expansion    | No heat exchange, gas does work on surroundings         | Temperature decreases | $Q = 0$       | Work done <b>by</b> the gas |
| Adiabatic Compression  | No heat exchange, work done on the gas                  | Temperature increases | $Q = 0$       | Work done <b>on</b> the gas |
| Isothermal Expansion   | Heat absorbed from surroundings to maintain temperature | Constant temperature  | Heat absorbed | Work done <b>by</b> the gas |
| Isothermal Compression | Heat released to surroundings to maintain temperature   | Constant temperature  | Heat released | Work done <b>on</b> the gas |